

PIERINA LOPEZ

Creative tech mind and architecture background

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Applied Computer Science student (EN/ES) with hands-on experience in web development, data dashboards, and smart home integration. Proactive and solutions-oriented, with a track record of delivering results under pressure.

My background spans architecture, hospitality, logistics, and telecoms, giving me an unusual mix of technical skills, customer relationship experience, and project coordination that I bring into every role I take on.



EDUCATION

2022 – 2026

Applied Computer Science
Thomas More University of
Applied Sciences, Belgium

2014 – 2019

Architecture and Urban
Planning
Cesar Vallejo University,
Peru

SOFT SKILLS

- Analytical problem-solving & debugging
- Teamwork & collaboration (Agile/Scrum, group projects)
- Adaptability & eagerness to learn new tools and tech
- Clear communication with technical & non-technical stakeholders
- Time management, ownership, and reliability on deliverables

CERTIFICATION

2016

AutoCAD Architecture
Certification
Cesar Vallejo University, Peru

VOLUNTEERING

VOLUNTEER BARISTA

The Big C, Belgium | Oct 2022 – Dec 2024

- Prepared and served hot/cold beverages while supporting a welcoming community space.
- Handled payments and maintained a clean, safe working area.

LANGUAGES

- Spanish - Native
- English - Fluent
- Dutch - Basic

TECHNICAL SKILLS

AI & Machine Learning

- Python (primary language, AI, IoT, and CV projects)
- LLM inference with Ollama (local) + RAG pipelines
- LangChain, LangGraph, ChromaDB (agent orchestration & retrieval)
- FastAPI + WebSockets (real-time AI streaming)
- Faster-Whisper (speech-to-text)
- TensorFlow / Keras (CNN, transfer learning — MobileNetV2)

Computer Vision

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- YOLOv8 (object detection & tracking)
- OpenCV (image processing, homography transforms)
- dlib (face recognition & landmark detection)
- Roboflow (dataset annotation & management)
- Streamlit (interactive CV demo UIs)

IoT & Embedded Systems

- Home Assistant — 26 automations (Mobilab & Care internship)
- AppDaemon (custom Python automation components)
- Raspberry Pi (GPIO, sensors, hardware migration Pi 3B → Pi 5)
- ESP32 + C++ (PlatformIO firmware, BMP280/BH1750 sensors)
- MQTT, Matter Hub, Alexa voice control
- Flask (local IoT APIs & web servers)

Web Development & Frameworks

- Laravel + Livewire (TALL stack — CRUD, auth, role & permission management)
- PHP + MySQL (schema design, migrations, queries)
- HTML5, CSS3, Bootstrap, Tailwind CSS
- JavaScript (basic), Alpine.js
- Figma (UX/UI prototyping), responsive design

Data & Analytics

- Qlik Sense (data modelling, dashboards, visualizations)
- Grafana (real-time monitoring dashboards)
- Streamlit (data apps & reporting UIs)
- Excel (intermediate, formulas, charts, filtering)

Tools & DevOps

- Git, GitHub, GitLab CI
- Docker (containerisation & deployment pipelines)
- Jira (Agile/Scrum project management)
- VS Code, AutoCAD, Revit
- Microsoft Office, Google Workspace

Internship

SMART HOME INTEGRATION INTERN

Research Group Mobilab & Care, Belgium | Feb 2026 – May 2026

- Designed and implemented a locally controlled smart home prototype on Home Assistant and Raspberry Pi, covering night-path lighting, bathroom safety monitoring, and help-mode escalation.
- Integrated Zigbee sensors, smart lighting, a Nuki smart lock, and voice control via Alexa using Home-Assistant-Matter-Hub, bridging multi-vendor devices through Zigbee2MQTT and MQTT.
- Built a Python analysis component (AppDaemon) that queries Home Assistant history, computes usage averages, and automatically updates weekday and weekend night-mode schedules.
- Applied academic research on ageing in place, lighting safety, and technology acceptance to ground every technical design decision in user-centred evidence.
- Produced full technical documentation including a system architecture, scenario-based test reports, an installation guide, and a user manual for future researchers and maintainers.

EMPLOYMENT HISTORY

BARTENDER / WAITER – Student Jobs

Post Restaurant, VOLT, Irish Pub – Belgium | Apr 2023 – Present

- Delivered high-quality service in fast-paced, high-end environments, managing simultaneous bar and table orders during peak hours.
- Prepared cocktails and beverages to customer specifications and house standards, contributing to consistently positive guest experiences.
- Monitored inventory, coordinated timely stock replenishment, and upheld hygiene and safety standards across all service areas.

OPERATIONS & WAREHOUSE ASSISTANT

Distrilog, Beyers, Greenyard – Belgium | Apr 2022 – Apr 2023

- Processed and prepared high volumes of products for shipment with accuracy and efficiency, consistently meeting daily throughput targets.
- Operated pallet stacking and wrapping equipment in compliance with warehouse health and safety regulations across multiple sites.
- Contributed to smooth logistics workflows by maintaining organised workstations and coordinating with team members to prevent bottlenecks.

CUSTOMER SERVICE & SALES REPRESENTATIVE

Teleperformance, Los Portales, Pet Center, Nuera Telecom – Peru & Remote | Jan 2013 – Aug 2021

- Managed high-volume customer inquiries and complaints across multiple industries, consistently achieving strong satisfaction scores.
- Applied upselling and cross-selling techniques to drive revenue and build long-term customer loyalty in competitive markets.
- Processed payments and maintained accurate records across CRM, POS, and billing platforms in fast-paced, multi-client environments.

ARCHITECTURE & URBAN PLANNING ASSISTANT

WAO, TEKUM, IMP, Municipality of Puente Piedra, Arch. Carmen Vásquez – Peru
| Jan 2015 – Mar 2020

- Produced architectural and urban-planning drawings in AutoCAD, Revit, and BIM workflows for public and private sector projects.
- Supported research, data analysis, and preparation of technical presentations for government authorities and private clients.
- Coordinated with multidisciplinary teams during site inspections, contributing to documentation, cost estimates, and compliance reporting.